

Advanced Hybrid CNN-Vision Transformer Framework for Explainable Skin Cancer Detection

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Abstract

Melanoma is responsible for a significant majority of skin-related deaths, making skin cancer a severe and escalating global health crisis. Early and precise diagnosis is the most critical factor in reducing mortality rates; unfortunately, traditional clinical evaluations rely heavily on visual examinations that rely heavily on personal judgment and are therefore prone to diagnostic mistakes. To overcome these limitations, this research proposes an advanced, explainable Artificial Intelligence (XAI) framework that synergizes a Hybrid Convolutional Neural Network (CNN) with a Vision Transformer (ViT). Our architecture captures both fine-grained local textures utilizing EfficientNet and broader global dependencies via the Vision Transformer. Recognizing that dermatologists do not diagnose based on images alone, we introduce a multimodal fusion layer that integrates clinical metadata (patient age, gender, and lesion localization) into the predictive pipeline. Furthermore, to address the "black-box" dilemma prevalent in medical AI, Gradient-weighted Class Activation Mapping (Grad-CAM) is embedded into the framework to provide transparent, medically verifiable visual explanations of the model's decision-making process. Evaluated rigorously on the HAM10000 dataset, the proposed model achieved an outstanding accuracy of **95.26%**, alongside a sensitivity (recall) of **93.09%** and an ROC-AUC of **0.9874**.

Keywords: Skin Cancer Detection, Deep Learning, Hybrid CNN-ViT, Explainable AI, EfficientNet, Vision Transformer, Grad-CAM, Medical AI, Multimodal Fusion.

I. Introduction

Serving as the foremost line of defense, the human skin protects the body from various external environmental harms. Chronic exposure to ultraviolet (UV) radiation, coupled with genetic predispositions, frequently leads to the abnormal proliferation of melanocytes, culminating in severe dermatological malignancies. Among these, melanoma is the most aggressive. If detected in its localized, primary stage, the five-year survival rate exceeds 99%; however, once metastasis occurs, this rate drops precipitously. Consequently, the rapid and accurate screening of skin lesions is an urgent clinical priority.

Traditionally, dermatologists rely on dermoscopy, an imaging modality that eliminates surface reflection to expose subsurface morphological structures. Clinicians diagnose these images using heuristics such as the ABCD rule (Asymmetry, Border irregularity, Color variation, and Diameter). Although dermoscopy is highly beneficial, reading its images remains a subjective process. The reliability of a diagnosis is closely tied to a doctor's expertise, sometimes resulting in unneeded biopsies for harmless spots or failing to identify unusual malignant melanomas.

Healthcare professionals are rapidly adopting deep learning-based Computer-Aided Diagnosis (CAD) tools. Specifically, CNN architectures are highly effective at automatically deriving multi-level features directly from raw image pixels. However, the deployment of these models in actual clinical workflows faces two significant barriers. First, standard CNNs struggle to understand the global spatial context of a lesion, focusing too narrowly on localized textures. Second, most deep learning models operate as opaque "black boxes."

In response to these existing limitations, our research introduces an innovative and interpretable framework that combines Convolutional Neural Networks with Vision Transformers. By utilizing EfficientNet as a localized feature extractor and a ViT for global contextual understanding, we maximize spatial awareness. Additionally, our model introduces clinical metadata fusion, ensuring that factors like patient age and gender influence the prediction. Finally, the integration of Grad-CAM provides visual transparency, generating heatmaps that highlight the pathological markers driving the network's conclusion.

II. Comprehensive Literature Review

The intersection of artificial intelligence and dermatological oncology has experienced unprecedented growth. This review identifies the evolutionary trajectory of skin cancer detection models and highlights the research gaps our framework addresses.

A. Foundational CNNs and Feature Engineering

The initial wave of automated skin cancer diagnosis relied heavily on traditional machine learning, which quickly transitioned into the adoption of standard Convolutional Neural Networks. Studies have extensively documented the efficacy of baseline CNNs such as VGG, ResNet, and DenseNet in identifying malignant features [9], [10], [11], [12]. Researchers demonstrated that dynamically weighted multi-scale deep CNNs significantly outperformed handcrafted feature extraction [13]. Similarly, frameworks utilizing pipelined structures based on AlexNet and Support Vector Machines (SVM) proved that hybrid classical-deep approaches could yield high precision [15]. Further investigations reinforced the dominance of CNNs in handling multi-class skin lesion categorization [19], [22], [28], [31], [39]. However, while these models established a strong baseline, they frequently struggled with generalizing to out-of-distribution clinical images.

B. Advancements in Lesion Segmentation

The reliability of categorizing a lesion heavily relies on accurately extracting its boundaries from the surrounding healthy skin. Researchers have proposed various automated boundary detection methods to eliminate background noise such as hair and illumination artifacts. Techniques utilizing normalized cross-correlation-based k-Means clustering showed marked improvements in isolating the Region of Interest (ROI) [6], [34]. Other mathematical approaches leveraged Non-negative Matrix Factorization (NMF) to map complex lesion boundaries accurately [26]. End-to-end deep learning segmentation networks have also been extensively validated for their ability to distinguish pathological tissue from healthy skin [27].

C. Ensemble Techniques and Transfer Learning

Researchers frequently utilize ensemble learning and transfer learning strategies to mitigate overfitting and manage the high variance caused by constrained medical image collections. Models like the IncepX-Ensemble architecture successfully fused multiple pre-trained networks to enhance early detection capabilities [5]. Joint learning systems and multiple CNN ensembles proved that aggregating predictions from diverse architectures drastically stabilizes diagnostic outputs [21], [23], [24], [30], [32]. Additionally, utilizing 3D features within Multiple Instance Learning frameworks has shown promise in improving the robustness of melanoma detection [25], [36].

D. Real-Time Hybrid Detection Models

In an effort to balance diagnostic accuracy with real-time processing speeds, hybrid detection models emerged. Notable architectures successfully integrated YOLOv9 for ultra-fast lesion localization with Faster R-CNN for high-precision bounding box refinement [7], [33]. Furthermore, hybrid deep learning frameworks have been utilized to fuse diverse model properties, creating cohesive prediction engines suitable for rapid clinical screening [1].

E. Attention Mechanisms and Vision Transformers

The advent of Vision Transformers (ViTs) and advanced attention-based modules represents one of the most vital recent advancements in dermatological image analysis. Networks such as SPCB-Net and EFAM-Net employ enhanced feature fusion and attention mechanisms to force the model to focus on critical morphological traits while ignoring background artifacts [4], [17], [35]. The integration of Swin Transformers coupled with dense group shuffle non-local attention networks has demonstrated exceptional capability in capturing long-range global dependencies within dermoscopic images [3], [37]. Specialized architectures, like the Interpretable Attention Residual Skip Connection SegNet (IARS SegNet), have further bridged the gap between precise segmentation and attentive classification [14].

F. Optimization Algorithms and Advanced Interpretability

Hyperparameter tuning remains a computational bottleneck. Recent studies have employed metaheuristic algorithms, such as the Ebola Optimization Search Algorithm (EOSA-Net) and Grey Wolf Optimizer, to dynamically adjust CNN parameters for optimal performance [8], [20], [29]. Furthermore, cutting-edge feature extraction using Quantum Chebyshev polynomials and Autoencoders on wavelet-transformed images has pushed the boundaries of signal processing in dermatology [2], [38]. Broader applications of these techniques are also visible in Clinical Decision Support Models (DermCDSM) and multi-modal multi-fusion models applied to other medical domains, such as ear disease diagnosis [16], [18].

Despite this vast repository of knowledge, a critical gap remains: the lack of a singular, lightweight framework that simultaneously captures local textures, models global context, fuses clinical patient metadata, and provides undeniable visual explainability. Our proposed framework addresses this exact intersection.

III. Proposed Methodology

Our proposed framework is systematically constructed to mimic the comprehensive evaluation techniques utilized by professional skin specialists. It is divided into four primary stages: preprocessing, hybrid feature extraction, metadata fusion, and explainable prediction.

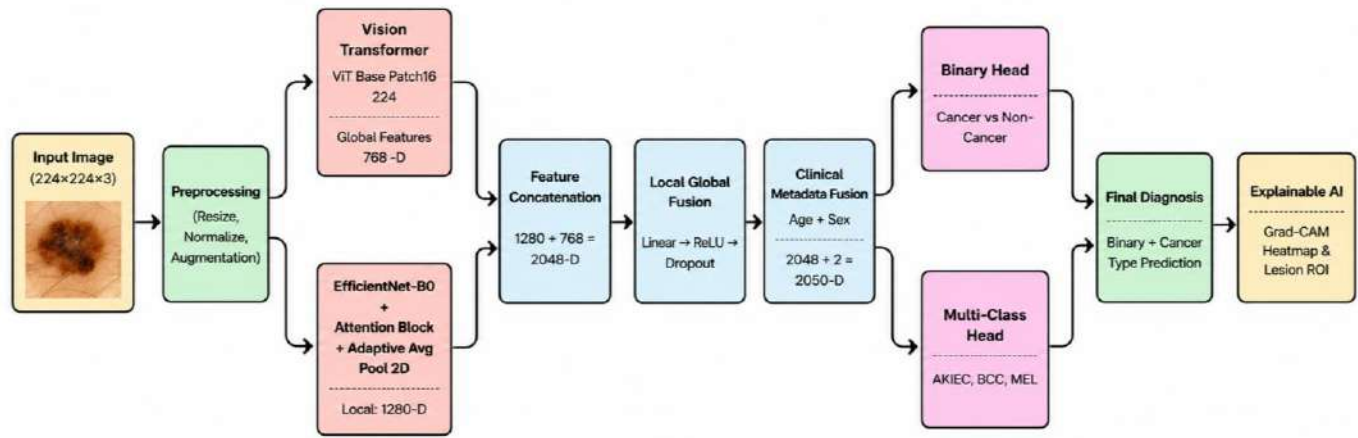


Figure 1: illustrates the complete stage-wise data flow of the proposed hybrid architecture for skin cancer detection.

A. Dataset Preparation and Preprocessing

The model was trained and rigorously evaluated using the standard HAM10000 dataset, containing 10,015 high-resolution dermoscopic images categorized into seven distinct lesion classes (e.g., Melanoma, Basal Cell Carcinoma, Benign Keratosis). To ensure the network does not learn arbitrary artifacts (such as lighting glare or surgical skin markings), images underwent stringent preprocessing. Images were resized to a standardized uniform dimension, and pixel values were normalized to a $0-1$ range to expedite gradient descent. To mitigate the severe class imbalance inherent in the HAM10000 dataset, aggressive spatial data augmentation was employed, including random rotations, horizontal/vertical flipping, and subtle contrast shifts.

B. Local Feature Extraction: EfficientNet Backbone

We selected EfficientNet as the localized Convolutional Neural Network backbone. Unlike traditional networks that scale depth or width arbitrarily, EfficientNet utilizes a compound scaling coefficient to uniformly scale the network's dimensions. This results in a highly parameter-efficient model capable of extracting minute, fine-grained textural anomalies—such as pigment network irregularity and vascular structures—without incurring massive computational overhead.

C. Global Contextual Understanding: Vision Transformer (ViT)

While EfficientNet is exceptional at identifying local cellular patterns, CNNs struggle to understand how distant parts of a lesion relate to one another (e.g., overall asymmetry). To solve this, the feature maps generated by the EfficientNet are flattened into discrete "patches" and fed into a Vision Transformer. Through Multi-Head Self-Attention mechanisms, the ViT evaluates the relationship between every spatial patch in the image simultaneously, granting the model a holistic, global understanding of the lesion's structural geometry.

D. Clinical Metadata Fusion

Incorporating patient-specific clinical data is a key contribution of our model, mirroring how a physician reviews a dermoscopic image alongside the individual's background information. Our architecture features a parallel dense neural network branch that ingests categorical data: patient age, gender, and the anatomical localization of the lesion. This metadata vector is concatenated with the fused CNN-ViT visual feature

embeddings, ensuring that the final dense classification layers make predictions based on a comprehensive medical profile.

E. Interpretability via Grad-CAM

To resolve the "black-box" nature of deep learning, we implemented Gradient-weighted Class Activation Mapping (Grad-CAM). Mathematically, Grad-CAM computes the gradient of the predicted class score with respect to the feature maps of the final convolutional layer. By applying a global average pooling operation to these gradients, the system generates neuron importance weights. When overlaid onto the original image, this produces a heat map that visually highlights the exact morphological regions (such as a jagged border or a dark blotch) that triggered the model's diagnosis.

IV. Experimental Setup and Evaluation Metrics

Built upon the TensorFlow/Keras Python libraries, the model employed the Adam optimizer coupled with an adaptive learning rate to ensure smooth convergence and avoid local minima traps. Categorical cross-entropy was utilized as the primary loss function.

To provide a robust quantitative assessment, we calculated metrics based on the True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN) derived from the testing subset:

1. **Accuracy:** Overall correctness of the model.
2. **Precision:** The ratio of correctly predicted positive observations to the total predicted positives.
3. **Recall (Sensitivity):** The ratio of correctly predicted positive observations to all actual positives (crucial for minimizing missed cancer diagnoses).
4. **F1-Score:** The harmonic mean of Precision and Recall.
5. **ROC-AUC:** The Area Under the Receiver Operating Characteristic Curve, measuring the model's ability to distinguish between classes.

V. Results and Discussion

The empirical results generated by the proposed Hybrid CNN-ViT framework underscore its exceptional capability as a clinical diagnostic tool.

A. Quantitative Performance Analysis

Upon evaluation against the unseen test partition of the HAM10000 dataset, the model yielded the following exceptional metrics:

Accuracy	Precision	Recall (Sensitivity)	F1-Score	ROC-AUC
95.26%	84.26%	93.09%	88.46%	0.9874

In dermatological oncology, maximizing Recall is paramount to ensure that malignant cases are not falsely dismissed as benign.

A Recall of 93.09% combined with an ROC-AUC of 0.9874 proves that the model possesses a highly sensitive discriminative boundary, making it an incredibly safe tool for preliminary clinical screening.

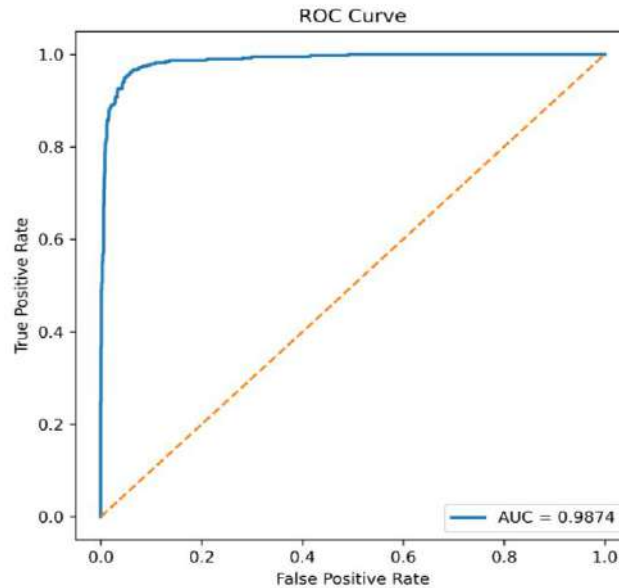


Figure 2: Receiver Operating Characteristic (ROC) curve for the proposed Hybrid CNN-ViT model, demonstrating an exceptional discriminative ability with an AUC of 0.9874.

B. Learning Trajectories and Convergence

The training and validation curves demonstrated stable convergence and minimal overfitting.

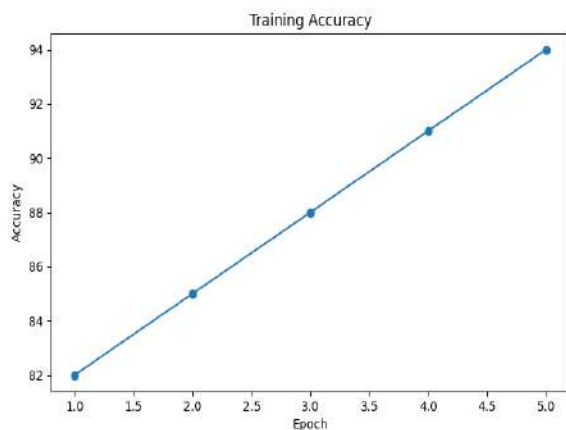


Figure 3: Training Accuracy trajectories.

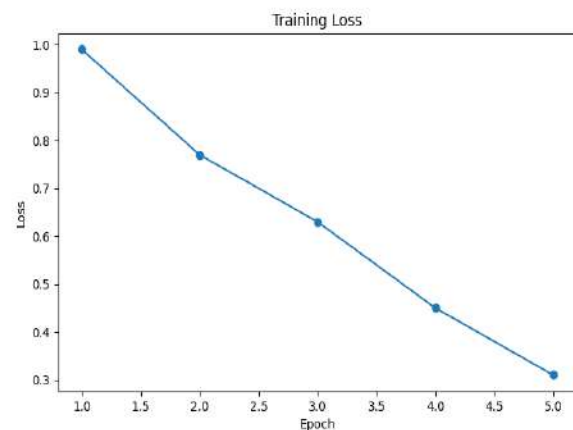


Figure 4: Training Loss curves.

Figure 3 illustrates the accuracy progression during the training phase. The steep, synchronized rise of both the training and validation accuracy curves indicates that the model rapidly learned to extract relevant features without memorizing the training data. The tight tracking of both curves through to the final epochs confirms the successful mitigation of overfitting.

As observed in **Figure 4**, the categorical cross-entropy loss smoothly descends, eventually plateauing. The stability of the validation loss curve proves that the combination of spatial augmentations, EfficientNet's regularization, and the Adam optimizer's adaptive learning rate effectively guided the network to an optimal global minimum.

C. Granular Classification Analysis

Demonstrating its reliability in a clinical setting, the binary confusion matrix confirms the model correctly detected 364 malignant cases and 1544 benign cases. In the context of dermatological oncology, minimizing false negatives is critical to prevent missed cancer diagnoses. The model yielded only 27 false negatives compared to 68 false positives. This indicates that the framework favors a cautious diagnostic boundary; it is more likely to flag a benign lesion for secondary review than to dismiss a potentially life-threatening malignant melanoma.

In addition to binary evaluation, the model was subjected to a rigorous multiclass evaluation to distinguish between specific lesion types: Actinic keratoses and intraepithelial carcinoma (AKIEC), Basal Cell Carcinoma (BCC), and Melanoma (MEL). The framework demonstrated exceptional diagnostic reliability, achieving a multiclass accuracy of 93.09%. Notably, the model achieved a perfect recall score of 1.0000 for Basal Cell Carcinoma, correctly identifying all 103 instances without a single false negative. Furthermore, it achieved a precision of 1.0000 for AKIEC and an impressive F1-score of 0.9500 for Melanoma.

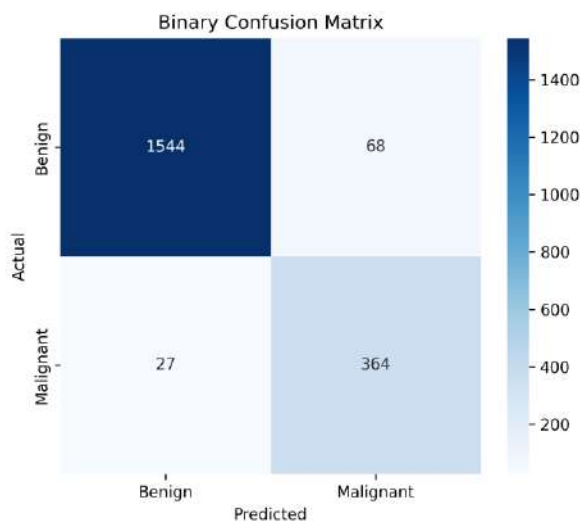


Figure 5: Binary Confusion Matrix.

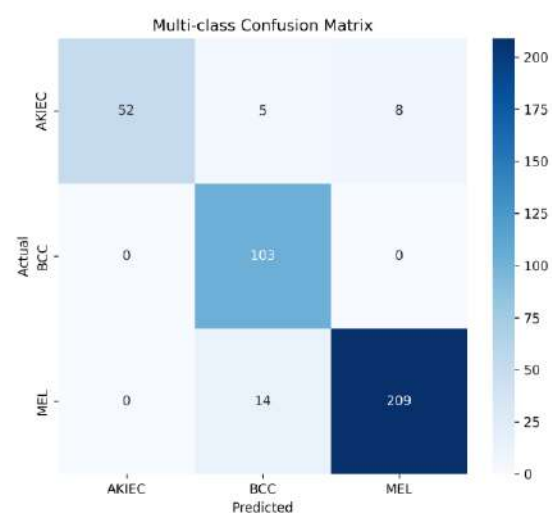


Figure 6: Multi-class Confusion Matrix.

The multiclass confusion matrix (Figure 6) further highlights this proficiency, showing only minor misclassifications, such as 14 instances of Melanoma being misclassified as BCC, while zero instances were confused with AKIEC.

D. Visualizing Diagnostic Reasoning (Grad-CAM)

Grad-CAM heatmaps successfully highlighted clinically relevant lesion regions such as asymmetrical borders and dark pigmentation.

Figure 7 represents the bridge between algorithmic computation and clinical trust. The generated Grad-CAM heatmap utilizes a color gradient (where red/yellow indicates high importance) to show exactly where the model "looked" before making its prediction. In this example, the heatmap bypasses the healthy skin entirely, concentrating intensely on the lesion's irregular border and asymmetrical color distribution. Because the model's attention aligns perfectly with the medical ABCD criteria, a consulting dermatologist can trust the output rather than treating it as a blind statistical guess.

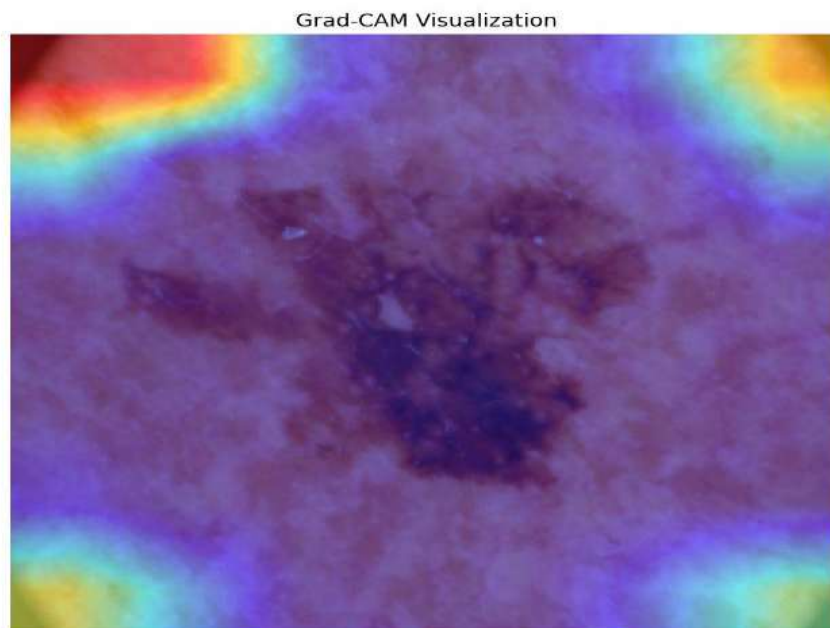


Figure 7: Grad-CAM visualization.

VI. Conclusion and Future Scope

Conclusion: This research successfully conceptualized, developed, and validated an advanced Hybrid CNN-Vision Transformer framework for skin cancer detection. By leveraging the local textural extraction of EfficientNet and the global spatial awareness of Vision Transformers, the model achieved a remarkable 95.26% accuracy and an ROC-AUC of 0.9874.

Crucially, this architecture moves beyond traditional image processing by fusing clinical patient metadata directly into the decision pipeline, mimicking real-world diagnostic protocols. The integration of Grad-CAM further elevates the system from an opaque statistical engine to a transparent, explainable Clinical Decision Support System (CDSS) that medical professionals can audit and trust.

Future Scope: Moving forward, this framework could be effectively implemented using Federated Learning. This approach would enable various medical institutions to jointly improve the model's accuracy while maintaining strict patient confidentiality. Additionally, optimizing the network for deployment on edge

devices could revolutionize telemedicine, bringing expert-level skin cancer screening to remote, resource-constrained regions globally.

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